

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Inventory Management System

Document Version: 4.0

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Project Type: Web-Based Inventory Management System

REVISION HISTORY

Version	Date	Description	Author
1.0	18/12/2025	Initial draft	Kalyani
2.0	20/12/2025	Project diagrams	Kalyani
3.0	26/12/2025	Finalised structure	Kalyani
3.5	12/01/2025	Filled in the info	Kalyani
4.0	27/01/2025	Finalized Doc	Kalyani

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete description of the functional and non-functional requirements of the **Inventory Management System**.

The purpose of this document is to:

- Define system behaviour clearly
- Serve as a reference for development and validation
- Act as a baseline for future enhancements
- Enable onboarding of future contributors

This document follows the **IEEE 830 standard** for SRS documentation.

1.2 Scope

The Inventory Management System is a web-based application designed to manage inventory items and track stock changes using a transaction-driven approach.

The system:

- Maintains inventory state
- Records all stock changes via immutable transactions
- Enforces role-based access control
- Provides a dashboard for operational visibility

Version 4 focuses on correctness, auditability, and maintainability.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
Item	An entity representing a resource in inventory
Transaction	A record representing stock IN or OUT
Admin	User with full system privileges
Staff	Authorized operational user
RBAC	Role-Based Access Control
ORM	Object Relational Mapper
UI	User Interface

1.4 References

- IEEE 830-1998 Software Requirements Specification
 - Django Official Documentation
 - Python Official Documentation
-

1.5 Overview

This document is organized into sections describing system context, features, interfaces, constraints, and future enhancements.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The system is a standalone web application developed using the Django framework. It follows a monolithic architecture suitable for small-to-medium scale inventory systems.

2.2 Product Functions

At a high level, the system provides the following functions:

- Inventory item management
 - Stock transaction tracking
 - Low-stock detection
 - Role-based access control
 - Dashboard analytics
-

2.3 User Classes and Characteristics

Administrator

- Technical user
- Full access to system
- Can manage inventory and users

Staff User

- Operational user
- Limited permissions
- Can create transactions and view inventory

Unauthorized User

- No access

2.4 Operating Environment

- Server: Platform independent
 - Runtime: Python 3.x
 - Framework: Django
 - Database: SQLite (development)
 - Client: Modern web browser
-

2.5 Design and Implementation Constraints

- Inventory updates must occur only through transactions
 - Direct stock editing is disallowed
 - Django authentication system must be used
 - Database must support atomic transactions
-

2.6 User Documentation

Could be found separately in the repository.

- *User Manual*
 - *Admin Guide*
-

2.7 Assumptions and Dependencies

- Users authenticate using Django authentication
 - System operates in a trusted internal environment
 - Database provides transactional integrity
-

3. SPECIFIC REQUIREMENTS

3.1 Inventory Item Management

Description:

The system shall allow administrators to create, view, update, and deactivate inventory items.

Functional Requirements:

- The system shall store item name, quantity, and minimum threshold
 - The system shall prevent permanent deletion of items
 - The system shall identify low-stock items
 - The system shall display current inventory state
-

3.2 Transaction Management

Description:

The system shall track inventory changes through IN and OUT transactions.

Functional Requirements:

- The system shall support IN and OUT transactions
 - The system shall prevent negative inventory
 - The system shall update inventory atomically
 - Transactions shall be immutable
-

3.3 Dashboard

Description:

The system shall provide an operational dashboard.

Functional Requirements:

- Display total items
- Display active items

- Display low-stock items
 - Display recent transactions
 - Visualize stock levels
-

3.4 Role-Based Access Control

Description:

The system shall restrict access based on user roles.

Functional Requirements:

- Admins shall have full access
 - Staff shall have limited access
 - Unauthorized users shall have no access
-

3.5 Authentication

Description:

The system shall authenticate users securely.

Functional Requirements:

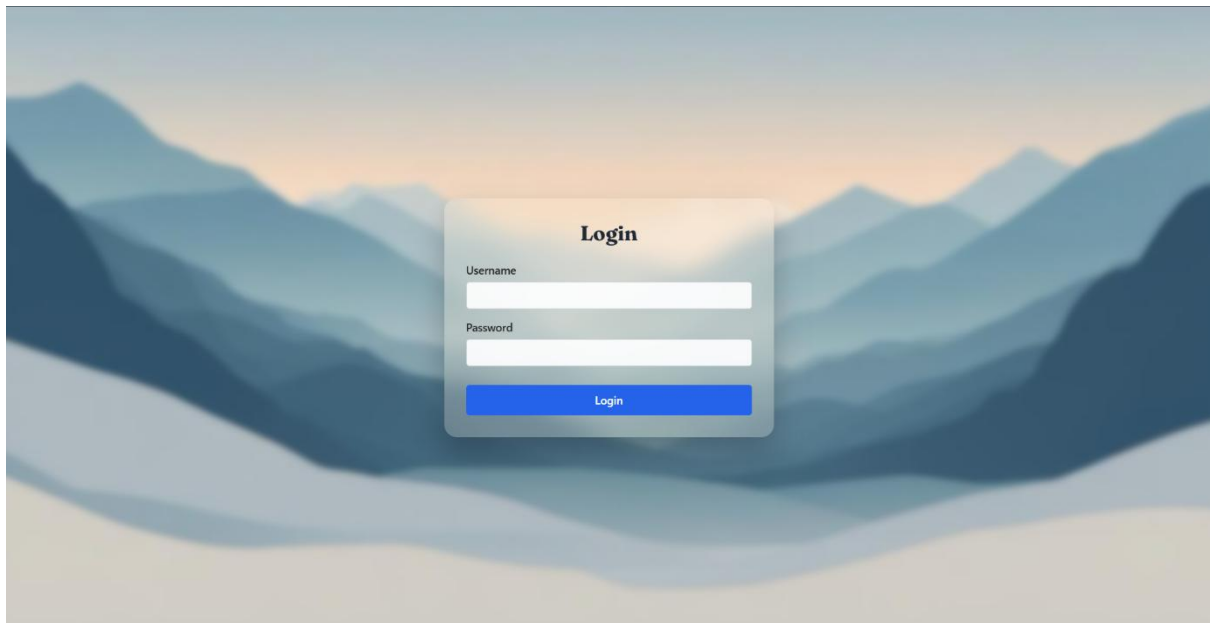
- Users shall authenticate before access
 - Secure session management shall be used
 - Logout functionality shall be provided
-

4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

- Web-based interface using Django templates
- Responsive design for standard screen sizes

Login



A login form is centered on a background image of misty mountains. The form is a semi-transparent white box with a blue 'Login' button. It contains fields for 'Username' and 'Password'.

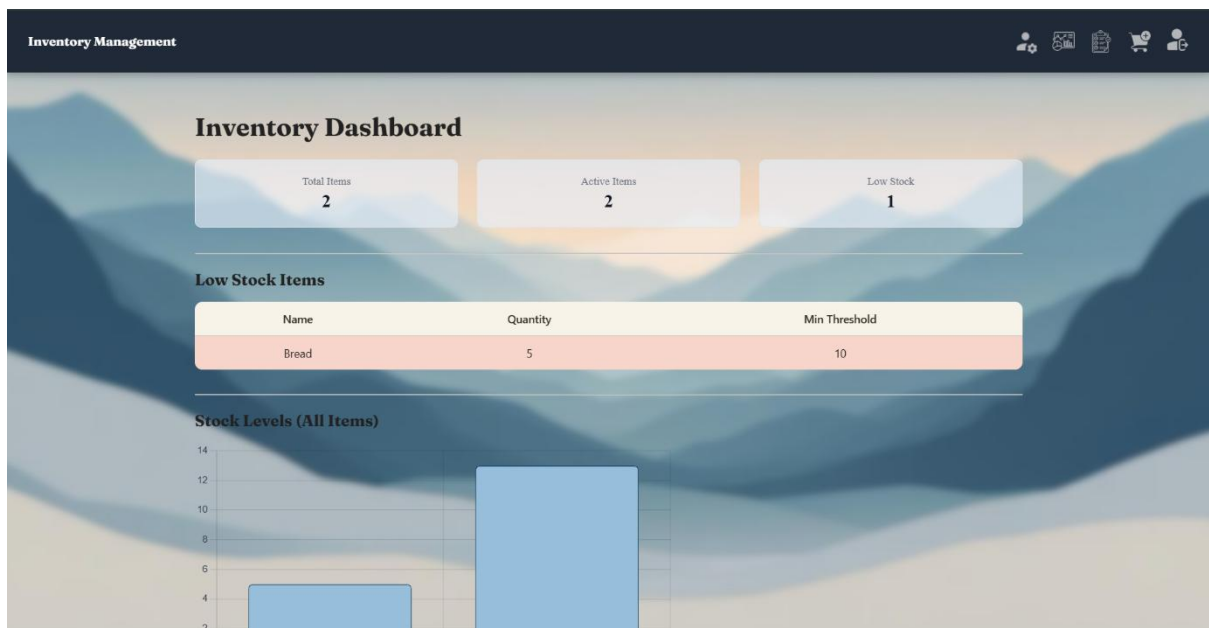
Login

Username

Password

Login

Dashboard



Dashboard



Item List

Inventory Management

Inventory Items

Name	Quantity	Minimum	Status
Bread	5	10	Low
Milk	13	2	OK

Transition History

Django administration

WELCOME, ADMIN_TEST. [VIEW SITE](#) / [CHANGE PASSWORD](#) / [LOG OUT](#)

Home > Inventory > Transactions

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

INVENTORY

Items + Add

Transactions

Select transaction to view

TRANSACTION TYPE	ITEM	QUANTITY	PERFORMED BY	CREATED AT
Out	Bread	20	staff_test	Jan. 5, 2026, 12:10 p.m.
In	Milk	5	admin_test	Dec. 26, 2025, 5:37 p.m.
Out	Milk	2	staff_test	Dec. 26, 2025, 4:50 p.m.
In	Milk	10	admin_test	Dec. 26, 2025, 4:49 p.m.
In	Milk	10	admin_test	Dec. 26, 2025, 4:47 p.m.
Out	Bread	10	kay	Dec. 26, 2025, 4:31 p.m.
In	Bread	20	kay	Dec. 26, 2025, 4:31 p.m.
Out	Bread	15	kay	Dec. 20, 2025, 3:52 p.m.
In	Bread	30	kay	Dec. 20, 2025, 3:46 p.m.
Out	Bread	20	kay	Dec. 20, 2025, 3:45 p.m.
In	Bread	10	kay	Dec. 20, 2025, 3:45 p.m.
Out	Bread	5	kay	Dec. 17, 2025, 4:46 p.m.
Out	Bread	10	kay	Dec. 17, 2025, 4:45 p.m.
In	Bread	5	kay	Dec. 17, 2025, 4:35 p.m.
Out	Bread	10	kay	Dec. 17, 2025, 4:33 p.m.

15 transactions

Transition

Inventory Management

New Inventory Transaction

Item

Select an Item

Transaction Type

In

Inbound adds stock. Outbound reduces stock.

Quantity

Remark (optional)

Submit Transaction

4.2 Hardware Interfaces

No specific hardware interfaces required.

4.3 Software Interfaces

- Django ORM
 - Relational database backend
-

4.4 Communication Interfaces

- HTTP / HTTPS
-

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance Requirements

- System shall respond within acceptable limits
 - Pagination shall be used for large datasets
-

5.2 Security Requirements

- Authentication required
- Role-based authorization enforced
- Data modification restricted

Action	Admin	Staff
Create item	✓	X
Update item	✓	X
Delete item	✓	X
View items	✓	✓
IN transaction	✓	✓
OUT transaction	✓	✓
View transactions	✓	✓
View low-stock	✓	✓
View analytics	✓	X

5.3 Reliability Requirements

- Atomic database transactions
 - Rollback on failure
 - Consistent data state
-

5.4 Maintainability Requirements

- Modular code structure
 - Clear separation of concerns
 - Easy extensibility
-

5.5 Scalability Requirements

- Support increasing inventory size
 - Replaceable database backend
-

6. DATA REQUIREMENTS

6.1 Item Data

- Name
 - Quantity
 - Minimum threshold
 - Active status
 - Timestamps
-

6.2 Transaction Data

- Item reference
 - Transaction type
 - Quantity
 - User reference
 - Timestamp
 - Remarks
-

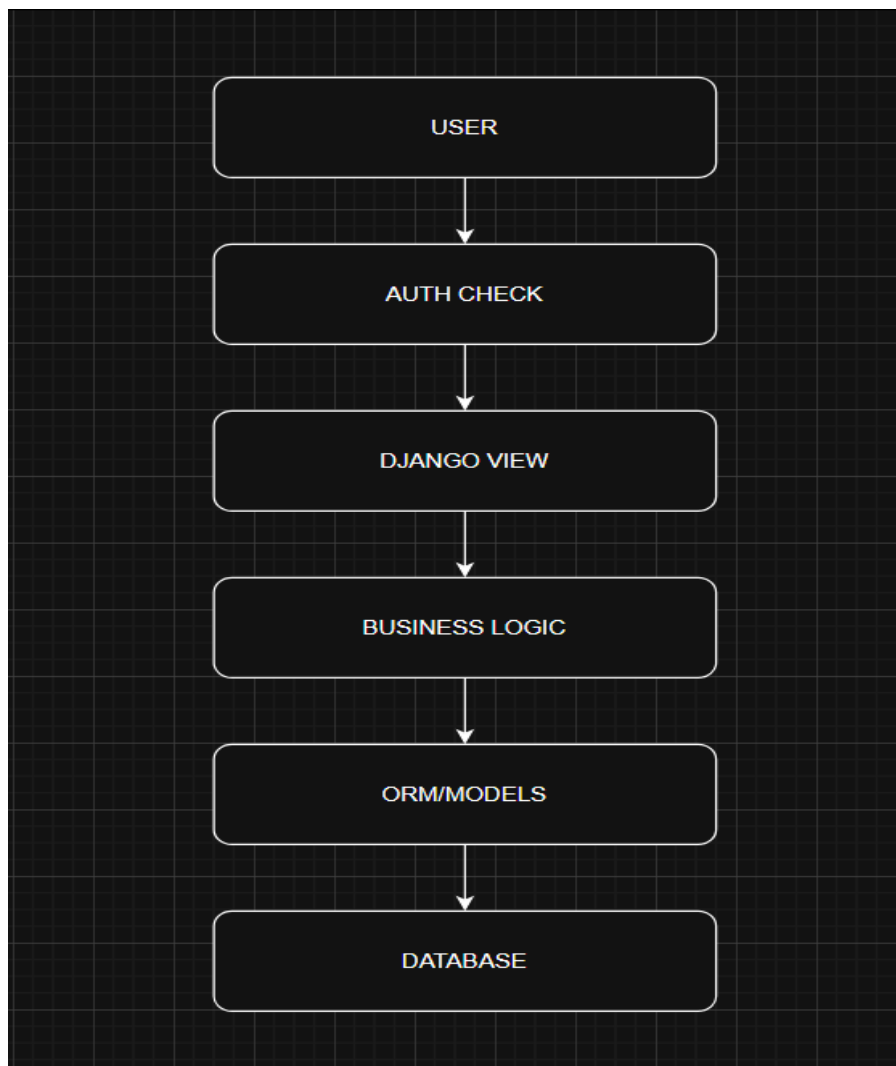
6.3 Data Access Boundaries

Action	Item	Transaction	User
Create Item	W	—	—
Update Item	W	—	—
Delete Item	W	—	—
View Items	R	—	—
IN Transaction	W	W	R
OUT Transaction	W	W	R
View Transactions	R	R	R
Low Stock View	R	—	—
Analytics	R	R	—

7. SYSTEM WORKFLOWS

Stock OUT Transaction Flow

1. User submits request
2. Authentication verification
3. Authorization check
4. Quantity validation
5. Stock availability verification
6. Atomic inventory update
7. Transaction recording
8. Commit or rollback



8. ASSUMPTIONS AND CONSTRAINTS

- Controlled user environment
 - Transaction-driven inventory model
 - Development database limitations
-

9. FUTURE ENHANCEMENTS

- Search and filtering
 - PostgreSQL integration
 - Deployment configuration
 - Reporting and exports
 - Automated testing
-

10. APPENDICES

Appendix A – Glossary

This glossary defines technical and domain-specific terms used throughout the Inventory Management System documentation.

The purpose of this section is to ensure clarity and a shared understanding for all readers, including developers, reviewers, and stakeholders.

Administrator (Admin)

A user with full system privileges. An administrator can manage inventory items, users, and system configuration, and has access to the Django admin panel.

Auditability

The ability of the system to provide a complete, traceable history of all inventory changes, including when a change occurred and who performed it.

Authentication

The process of verifying a user's identity before granting access to the system, typically through a username and password.

Authorization

The process of determining what actions an authenticated user is permitted to perform within the system, based on their assigned role.

Business Rules

Logical constraints and conditions enforced by the system to ensure correct and valid operation, such as preventing negative inventory.

Dashboard

A user interface that presents summarized inventory information, including item counts, low-stock alerts, recent transactions, and visual charts.

Data Integrity

The assurance that data remains accurate, consistent, and reliable throughout its lifecycle within the system.

Django Admin Panel

A built-in administrative interface provided by the Django framework, used for managing application data and users in a secure manner.

Immutable

A property of data that cannot be modified or deleted once created. In this system, transactions are immutable.

Inventory Item (Item)

An entity representing a physical or logical resource being tracked in the inventory system, such as a product or material.

Low-Stock Item

An inventory item whose current quantity is less than or equal to its defined minimum threshold.

Minimum Threshold

The minimum acceptable quantity for an inventory item, used to detect and flag low-stock conditions.

Pagination

A technique used to divide large datasets into smaller pages to improve performance and usability.

Role-Based Access Control (RBAC)

A security model in which system permissions are assigned based on user roles, such as Admin or Staff.

Staff User

An authorized operational user who can view inventory data and create stock transactions but cannot manage system configuration.

Stock IN Transaction

A transaction that increases the quantity of an inventory item, representing the addition of stock.

Stock OUT Transaction

A transaction that decreases the quantity of an inventory item, representing the removal or usage of stock.

Transaction

A record representing a single, atomic inventory change event. Each transaction includes the item affected, type, quantity, user, and timestamp.

Transaction History

The complete, chronological record of all transactions performed within the system.

Transactional Integrity

The assurance that inventory updates occur atomically and consistently, with either all changes applied or none applied.

Unauthorized User

A user who is not authenticated or does not have permission to access system features.

User Session

A temporary state maintained by the system to track an authenticated user's interaction during a login period.

Visualization

Graphical representation of data, such as charts displaying stock levels or inventory trends.

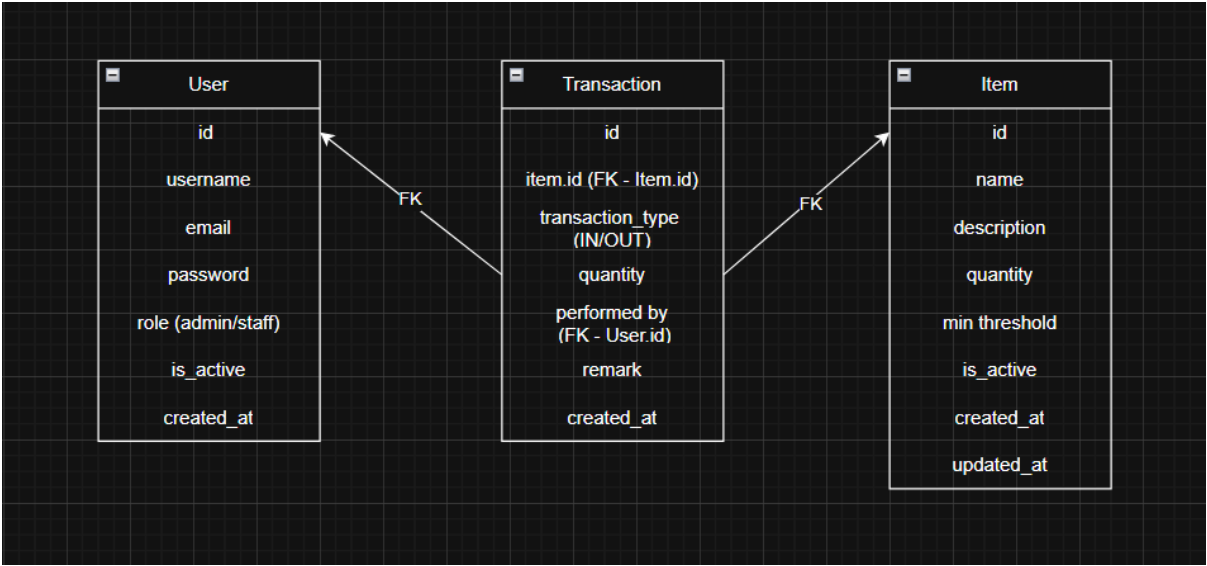
Web Application

A software application accessed through a web browser, requiring no local installation by the user.

Appendix B – Use Case Diagrams



Appendix C – ER Diagram



Action	Admin	Staff
Create item	✓	X
Update item	✓	X
Delete item	✓	X
View items	✓	✓
IN transaction	✓	✓
OUT transaction	✓	✓
View transactions	✓	✓
View low-stock	✓	✓
View analytics	✓	X

11. CONCLUSION

This document defines the complete IEEE 830–compliant Software Requirements Specification for the Inventory Management System. The system is designed with a focus on correctness, auditability, and maintainability, providing a strong foundation for future production deployment.